

E-Commerce Sales Performance Analysis (2009–2011)

1. Executive Summary

This report analyzes three years of transactional data (2009–2011) from the `online_retail_II.xlsx` dataset, migrated into a normalized PostgreSQL database. The analysis reveals a business heavily driven by the **United Kingdom**, with extreme seasonality peaking in **Q4**. While a single product category (White Hanging Heart T-Light Holder) dominated volume in the first two years, 2011 saw a shift toward high-volume disposable goods (Jumbo Bags). Notably, a significant portion of revenue is attributed to unspecified customer accounts (NaN), suggesting a large segment of guest checkout or unregistered transactions.

Key Findings:

- **Record Breaking Peak: December 2010** was the highest-performing month in history with **\$256.7M** in revenue.
- **Product Shift:** The best-selling product by volume shifted from decorative items (2009–2010) to practical packaging (2011), where the *Jumbo Bag Red Retrosport* sold nearly **900,000 units**.
- **Hero Products in Peak Months:** During the busiest months, specific "hero" products drove massive volume, such as the *Pack of 72 Retrosport Cake Cases* in Dec 2010 (208k units).
- **Customer Concentration:** The **United Kingdom** dominates the top spender list. The top individual spender in 2010 (Customer 14646, Netherlands) contributed **\$11.9M**.
- **Order Value:** The average order value (AOV) is \$18.07, while the most common order (MOV) is \$15, indicating that while the average is pulled up by larger orders, the typical purchase is smaller.
- **Scale:** The database contains 5,943 unique customers and 1,067,371 transactions.
- **Data Standardization:** The country label "EIRE" was identified as the Irish name for Ireland and was standardized in the database to ensure consistent regional reporting.

Recommendation: Immediate focus is required on identifying the NaN customer segment to capture lost loyalty data. Inventory planning must prioritize the "Hero Products" identified for Q4, specifically the *Retrosport* and *Heart T-Light* lines.

Focus on converting guest checkouts and use the \$15 MOV as a benchmark for upselling (e.g., "Spend \$5 more for free shipping").

2. Introduction

Business Scenario Management requires a strategic review of performance to guide planning for the upcoming fiscal year. The primary objective is to understand the core drivers of the business: product popularity, customer value, regional strength, and seasonal timing.

Objective This project demonstrates end-to-end data analytics by:

1. **Extracting & Loading** raw data from Excel into a structured PostgreSQL database.
2. **Designing** a normalized schema (invoice, products, customers).
3. **Calculating KPIs** to answer:
 - Which months generated the highest revenue?
 - Which products sold the most units annually and during peak months?
 - Which customers and regions contribute the most value?

Scope The analysis covers **2009–2011**, utilizing in-database SQL analytics to process millions of transaction rows.

3. Methodology

Phase 1: Database Design & Migration Raw data from online_retail_II.xlsx was migrated into three normalized tables:

- invoice: Fact table (Transaction metrics).
- products: Dimension table (Item descriptions).
- customers: Dimension table (Country, CustomerID).

Phase 2: Relational Structure We established **One-to-Many relationships** linking invoice to products and customers via foreign keys. This allowed us to:

- Join transactional volumes with product descriptions.
- Attribute revenue to specific countries and customer IDs.
- Aggregate data across dimensions without redundancy.

Phase 3: In-Database Analysis and Cleaning

- **Data Standardization:** An SQL UPDATE command was executed to standardize the country name 'EIRE' to 'Ireland' for consistent reporting.
- **Complex Queries:** Utilized JOIN, GROUP BY, and Window Functions (ROW_NUMBER, RANK) to identify top performers per year and per peak month.

4. Findings & Analysis

4.1 Seasonal Revenue Trends

Sales are intensely seasonal, with the highest revenue always occurring in **November or December**.

- **2010 (Peak Year):** December generated **\$256,663,166.72**, the highest single-month revenue in the dataset.
- **2011:** Peak shifted slightly to November with **\$150,292,728.33**.

- **2009:** December led with **\$77,475,021.00**.
- *Insight:* The business more than tripled its peak monthly revenue from 2009 to 2010.

4.2 Product Performance

Annual Volume Leaders:

- **2009 & 2010:** The **White Hanging Heart T-Light Holder** was the dominant product, selling **95,474** and **116,048** units respectively.
- **2011:** A shift in consumer behavior occurred; the **Jumbo Bag Red Retrospot** became the volume leader with a massive **890,988** units sold.

"Hero" Products in Peak Months: During the highest revenue months, specific items drove the surge:

- **Dec 2010:** *Pack of 72 Retrospot Cake Cases* (208,601 units; \$619k revenue).
- **Nov 2011:** *Rabbit Night Light* (206,645 units; \$548k revenue).
- **Dec 2009:** *White Hanging Heart T-Light Holder* (95,474 units; \$265k revenue).

4.3 Customer & Regional Insights

The **United Kingdom** is the undisputed primary market, housing the top spending accounts every year.

Top Spenders by Year:

Year	Rank	Customer ID	Country	Total Spent
2009	1	NaN	United Kingdom	\$57,857,611
	2	13694	United Kingdom	\$1,601,830
	3	15823	United Kingdom	\$580,915
2010	1	NaN	United Kingdom	\$412,790,886
	2	14646	Netherlands	\$11,880,468
	3	14156	Ireland	\$9,522,222
2011	1	NaN	United Kingdom	\$347,073,031

2	14096	United Kingdom	\$28,532,583
3	14646	Netherlands	\$17,534,913

- **Critical Observation:** The top "customer" is listed as **NaN** (Unknown/Guest) for all three years, accounting for hundreds of millions in revenue. This indicates a massive volume of sales where customer data was not captured or linked correctly.
- **Identified VIPs:** Customer **14646 (Netherlands)** is a consistent high-value client, appearing in the top 3 for both 2010 and 2011.

4.4 Average vs. Modal Order Value

- **Average Order Value (AOV): \$18.07**
- **Modal Order Value (MOV): \$15.00**
- **Interpretation:** The MOV being lower than the AOV indicates that a few large orders are pulling the average up. The most frequent transaction is for \$15, suggesting a core customer behavior. This \$15 point is a strategic opportunity for upselling and cross-selling.

4.5 Scale of Operations

- **Total Unique Customers: 5,943**
- **Total Transactions: 1,067,371**
- **Insight:** The high transaction count relative to the number of customers confirms a significant volume of guest checkouts (NaN), as many customers do not make multiple tracked purchases.

5. Conclusions & Recommendations

Conclusions The business has seen explosive growth, particularly in 2010. However, reliance on the UK market and "Guest" (NaN) transactions presents both a strength and a risk. The product mix is dynamic, shifting from decor to high-volume consumables.

Recommendations

1. **Data Integrity Audit:** Investigate the NaN customer IDs immediately. Capturing this data could unlock hundreds of millions in identifiable customer value for loyalty programs. Implement a post-purchase email capture or loyalty program to convert NaN guest checkouts into identifiable customers.

2. **Upsell at \$15 Threshold:** Use the MOV of \$15 to design promotions. For example, offer free shipping on orders over \$20 or a small gift for orders over \$25.
3. **Inventory Strategy:**
 - Secure massive stock of **Jumbo Bags** and **Retrosport** items for Q4 2012, given the 2011 surge.
 - Maintain steady supply of **White Hanging Heart T-Lights** as a perennial favorite.
4. **VIP Retention:** Launch a dedicated account management program for **Customer 14646 (Netherlands)** and other identified top spenders to prevent churn.
5. **Market Expansion:** While the UK is dominant, the strong showing from the Netherlands and EIRE suggests potential for targeted marketing campaigns in mainland Europe.

6. Appendix

- **A.1 Data Cleaning:** Executed UPDATE customers SET country = 'Ireland' WHERE country = 'EIRE'; to standardize regional names.
- **A.2 Top Month**
Query: Used EXTRACT(MONTH) and SUM(quantity*price) with ORDER BY desc.
- **A.3 Product Ranking:** Used ROW_NUMBER() OVER (PARTITION BY year ORDER BY quantity DESC).
- **A.4 Customer Aggregation:** Grouped by customerID and country with SUM aggregation.